

Worksheet 4D: Transcendental Functions via Series — Solutions

AIML Foundations Mathematics
Dublin and Dún Laoghaire ETB
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Instructor Copy — Not for Distribution

Note to instructor: Answers below reflect the expected responses. Where a question asks for reasoning or explanation, sample responses are given. Accept any mathematically equivalent form unless the question specifies a particular form.

Part A

1. $1 + x + \frac{x^2}{2} + \frac{x^3}{6} + \frac{x^4}{24} + \frac{x^5}{120}$
2. 1
3. ≈ 2.7167 (actual $e \approx 2.7183$)
4. ≈ 1.1052
5. $\frac{x^2}{2}, \frac{x^3}{6}, \frac{x^4}{24}$
6. $1 + x + \frac{x^2}{2} + \frac{x^3}{6} + \dots$
7. It's the same series! $\frac{d}{dx}e^x = e^x$
8. $e^x + C$
9. $2e^{2x}$
10. $2xe^{x^2}$
11. $\frac{1}{3}e^{3x} + C$
12. $e^{x^2} + C$

Part B

13. odd
14. even
15. 0
16. 1
17. ≈ 0.09983
18. ≈ 0.99500
19. $\frac{x^4}{24}, -\frac{x^6}{720}$
20. $\frac{x^4}{24} - \frac{x^6}{720}$

21. It's the cosine series!
22. $\frac{x^3}{6}, -\frac{x^5}{120}$
23. $\frac{x^3}{6} - \frac{x^5}{120}$
24. $-\sin(x)$
25. $\sin(x)$
26. 4
27. $-\sin(x)$ (since $100 = 4 \times 25$, back to start)
28. $\sin(x) + C$
29. $-\cos(x) + C$

Part C

30. $\ln|x| + C$
31. $\ln|x| + C$
32. $\frac{1}{x}$
33. $\frac{2}{x}$ (or use $\ln(x^2) = 2\ln(x)$)
34. $\frac{2x}{x^2+1}$
35. $\ln(x^2 + 1) + C$
36. $\frac{1}{2} \ln|2x + 3| + C$
37. reciprocals
38. $\frac{d}{dx} \ln(x) = \frac{1}{e^{\ln(x)}} = \frac{1}{x}$

Part D

39. $\frac{d}{dx}[e^x(x-1)] = e^x(x-1) + e^x = e^x \cdot x$
40. $x \sin(x) + \cos(x) + C$
41. $-x \cos(x) + \sin(x) + C$
42. $e^x(x^2 - 2x + 2) + C$
43. $\frac{e^x(\sin x - \cos x)}{2} + C$
44. $x \ln(x) - x + C$
45. $\ln(x) + 1 - 1 = \ln(x)$
46. $\frac{x^2 \ln(x)}{2} - \frac{x^2}{4} + C$

Part E

- 47. See work
- 48. Real: $\cos(x)$; Imaginary: $\sin(x)$
- 49. $-1 + 0i = -1$
- 50. rotation/circular motion

Part F

- 51. $e^x, \frac{1}{x}, \cos(x), -\sin(x)$
 - 52. $e^x + C, \ln|x| + C, -\cos(x) + C, \sin(x) + C$
 - 53. series/Taylor
 - 54. infinite
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